

Ceramics Lecture II: Porcelains, Decorative Techniques

Monticello FS 2026

Decorative Techniques (tips to ID):

Handpainting

Underglaze

Painted decoration applied to the clay body beneath the glaze.

Look for:

- Evidence of brush strokes
- Color diffusion (parts of the design may be darker or lighter in color depending on the thickness or amount of paint on the brush when applied)
- Crazeing in the glaze on top of decoration
- If decoration extends up to the fractured edge of a sherd, you should be able to see the paint sitting under the glaze in cross section (tip: use a derm lite!)

Remember, certain color palettes can be temporally diagnostic!

- Polychrome Warm
 - “Warm” hues: soft yellows, olive greens, browns, occasional blues
 - Botanical and floral motifs extremely common; neoclassical elements
 - *Predominantly* found on Pearlware, c. 1795-1830
- Polychrome Cool
 - “Cool” hues: intense greens, blues, light pinks, magentas (think chromatic, almost neon)
 - Botanical and floral elements still dominate, but designs tend to be less precise, thicker brush strokes
 - Predominantly seen on Whiteware and Ironstone, c. 1829-1920

Overglaze

Painted decoration applied over the glaze after a vessel has been fired. A second kiln firing at a lower temperature is typically required for the design to adhere to the surface.

Look for:

- Texture (can usually feel decoration over glaze)
- Lines/brush strokes tend to be crisper
- Overglaze degrades easily, so any evidence of deterioration is key
 - Flaking paint

- Ghosting (i.e., the “phantom” impression of decoration that remains after pigment has flaked off)
 - Dull, discolored, or rough spots in the decoration
- Gilding and luster decoration always over-glaze
 - Luster: thin metallic film/glaze applied as over-glaze finish
- If decoration extends up to the fractured edge of a sherd, you should be able to see the paint sitting on top of the glaze in cross section (tip: use a derm lite!)
- Overglaze decoration appears on many ware types, but is particularly common on porcelains

Slip

Slip: liquid clay (clay + water), often with added colorant.

Hand-slipped

Slip decoration applied by hand under the glaze.*

Look for:

- Texture (can usually feel trails of slip under glaze)
- Slip visible under glaze in cross section along broken edges
- Opacity: slip decoration is opaque and obscures underlying paste color (whereas the un-slipped glazed surface is typically translucent)
- Hand-slipped decoration mostly on coarse earthenwares (e.g., Staffordshire/North Midlands)
- “Trailed” decoration common

*Note: Slipping also occurs on *unglazed* indigenous CEWs/Colonoware (not at Monticello, to our knowledge).

Factory Made Slipware

Slip decoration applied with a lathe using a wide variety of decorative techniques, slip colors. Sometimes referred to as “dipped” ware.

Note: Multiple, specific decorative techniques commonly utilized on same vessel.

Look for:

- Regular, often geometric designs in bands encircling vessel very common
- Shallow, regular, often geometric patterns cut into surface (engine-turned)
- Often utilizes multiple colors of slip
- Most commonly seen on refined earthenwares
- Slip visible under glaze in cross section along broken edges
- Texture, opacity

Commonly seen dec techs on Factory-Made Slipware:

FYI: For your exam, you only need to know the overarching decoration type (or “genre,” in DAACS speak): i.e., “**Factory Made Slipware.**” However, the following details may help you to ID.

Solid and Banded Slip

Solid backdrop and/or bands of multi-colored slip are applied via blowing bottle or brush while leather-hard vessel is turned on the lathe.

Variegated (marbled)

Multiple slip colors applied to surface while leather-hard vessel is turned on the lathe. Colors run and swirl together to create a marbled effect.

Dendritic (“Mocha”)

Dendritic (“tree-like”) designs applied to slipped surface of leather-hard vessel. Potter uses a brush to distribute small amounts of an acidic solution onto the still-wet slipped surface, causing a reaction that spreads and expands the pattern. Acidic solution typically concocted of various unsavory ingredients, including urine, tobacco juice, hops, ground iron).

Multi-chambered slip

Multiple slip colors applied in various patterns using a multi-chambered slip cup, which allowed for 3-4 different colors to be distributed at once. “Cat’s Eye,” and “Common Cable” popular designs produced via this method.

Engine-turned decoration:

Combination of colored slip decoration and regular, geometric patterns cut into surface using an engine-turning lathe.

Use of the engine-turning lathe for ceramic production begins second half of the 18th century.

Two methods:

- Solid color or bands of colored slip applied to leather hard surface and allowed to set. Then, vessel is attached to engine-turning lathe equipped with fixed blades and other mechanisms to cut designs into the slipped surface and expose underlying body color.
- Vessel is first turned on the engine-turning lathe and regular patterns are cut directly into the leather-hard paste. Colored slip is then applied and allowed to set. vessel is then re-attached to the lathe and excess slip is shaved away to expose the cut pattern and the darker, inlaid color in the recessed areas.

Transfer Printing

Decorative technique in which a design engraved on a copper plate is transferred to the vessel surface using tissue paper or a glue bat and may be applied over or under the glaze.

Most common first half of 19th century.

Look for:

- Intricate designs, often covering large swathe of total surface area (relative to handpainting)
- Stippled or “pixelated” appearance, especially on later wares
- Field dots (clusters of small circles fill in negative space) on some examples
- *Lack* of evidence for brush strokes, little color diffusion or variety in hue
- Seen *predominantly* on refined earthenwares (e.g., CW, PW, WW, Ironstone)
- Underglaze much more common historically
- For overglaze, look for a noticeable texture over the glaze, evidence of deterioration

Overglaze transfer printing

- Developed c. 1750s
- Method: (1) Design transferred in oil from copper plate onto thin sheet or “bat” of glue and placed on the *glazed* (glost fired) ceramic surface. (2) Colored enamels (typically in powder form) applied to design (excess scraped off). (3) Design set by firing vessel once again at a lower temperature.
- Often on Creamwares;* muted/darker colors common
 - *also see (rarely) on later 19th century wares

Underglaze transfer printing

- Developed c. 1780s
- Method: (1) Ink/paint applied to copper plate design and pressed onto a sheet of tissue paper. (2) Design is transferred by pressing paper onto the unglazed (bisque fired), porous ceramic surface to. (3) Vessel then lightly fired to set the design, then glazed and fired a final time (glost firing)
- Cobalt blue most common TP color, but various other colors also used (especially later in the 19th century)

Molding

Raised/recessed designs formed in vessel mold or molded separately and applied to surface (sprig molding).

- Sprig molding common on German stoneware, dry-bodied stonewares (e.g., Black Basalt)
- Royal, Queen’s, Feather Edge on White Salt Glaze, Whieldon, and Creamware
- Shell-Edged
 - Blue/green varieties overwhelmingly the most common

- Most ubiquitous of all molded edge decorations (cheapest painted tableware option in 19th century)
- Refined earthenwares: CW, PW, WW, Ironstone
- Can use stylistic details to date more precisely (overall, designs get less complex, more sloppily executed over time)

Incising

Design cut or scratched into the wet or leather-hard ceramic body using a hard tool, either by hand or mechanically on a lathe.

- Commonly seen on indigenous CEWs/Colonoware (uncommon at Monticello)
- Also, Scratch Blue (White Salt Glaze), Westerwald/Rhenish
- Dry bodied stonewares, factory made slipwares often incised via engine-turning lathe

Impressing

Surface treatment or decoration applied by using a secondary material to press a pattern or texture into the ceramic surface prior to firing. Most commonly associated with Indigenous CEWs. Cordage, fabric, and other organic materials commonly used to create pattern, often applied via wrapped paddle.

- Cord marking **overwhelming** most common type of impressing found on Monticello CEWs

Extras: Other Porcelains

FYI just for your reference..since we had to skip over this. With the exception of Bone China, you're unlikely to see these on the mountaintop, & don't panic—you're not going to be tested on this (to my knowledge).*

**I suppose unless something crazy happens and we end up finding a billion sherds of bone china over the next 5 weeks??—in which case, you'll be able to recognize it*

English Bone China

Of the “other” porcelains, this is the one you are *most* likely to encounter during this field school.

- Date: c. 1794—present (**true** manufacturing date range)
- Fired at a relatively low temperature for porcelain (around 1200C+)
 - This compensated by the addition of bone ash, which acts as a flux to lower the melting temperature
- Lead glazed
- **Look for:**
 - Dense, translucent paste (as all porcelains), but kind of “crunchy” (?)
 - Surfaces often have ivory or creamy tint, particularly in archaeological contexts (looks kind of like Creamware at first)

- Paste and surfaces *slightly* more prone to staining (relative to hard paste)
- Lead glaze subject to crazing (unlike hard paste)
- Forms often thin-walled, teawares
- Overglaze and molded decoration common

English* Soft Paste

- Date: c. 1745-1795 (**true** manufacturing date range)
- Fired at a relatively low temperature relative to other porcelains (around 1100-1250 C)
 - 2 firings a must → second firing at lower temperature after glaze is applied (lead can't withstand heat of glaze or "glost" firing)
- Lead glazed
- **Look for:**
 - Relatively dense, translucent paste (not quite at the level of Bone China or hard paste, but distinct from refined earthenwares)
 - "Chalky" paste texture (almost like a refined earthenware, but not quite)
 - More prone to staining (relative to hard paste)
 - Surfaces may have a slightly greenish or greyish-green tint
 - Lead glaze subject to crazing (unlike hard paste)
 - Handpainted blue decoration most common in British colonial contexts

*Most common, but soft paste porcelain is manufactured in other parts of 18th c. Europe and (very briefly) colonial America.

Japanese Porcelain

- Date: c. 17th century*—present (**true** manufacturing date range)
 - **Very rarely imported to America before c. 1870*
- **Hard paste** body → dense, vitrified, gray/white paste
- Feldspathic/alkaline glaze
- Wide variety of decorative techniques
- **Look for:**
 - Paste and glaze characteristics (unfortunately) very similar to CP
 - Decoration is important!
 - **Transfer print** (you will **not** see transfer print on Chinese Porcelain)
 - Handpainted overglaze with luster common on later examples
 - In general, handpainted motifs tend to be less sharply executed relative to CP examples
 - Earlier vessels tend to be thicker-bodied, with bulkier forms
 - Glaze on early examples often thicker relative to CP
 - Context! You're simply much more likely to encounter Japanese porcelain on sites with later 19th century and 20th century occupations and/or in western U.S. archaeological contexts.

Disclaimers (again):

Lots of caveats and exceptions here, but don't panic. Yes, many complexities (wise to keep in mind, esp. if you continue in archaeology!).

That being said, in general, and especially in the context of this field school (i.e., *your exam*).. when you hear hoofbeats: think horse, *not* zebra!

Again, **all dates are context dependent!**

Helpful Resources!

Diagnostic Artifacts in Maryland (Jefferson Patterson Park and Museum):

[Diagnostic Artifacts in Maryland](https://apps.jefpat.maryland.gov/diagnostic/index-Ceramics.html) (<https://apps.jefpat.maryland.gov/diagnostic/index-Ceramics.html>)

Ceramics in America (Chipstone Foundation):

[The Chipstone Foundation](https://www.chipstone.org/publications.php/2/Ceramics-in-America) (<https://www.chipstone.org/publications.php/2/Ceramics-in-America>)